becoming a catastrophic problem that quickly spreads across the national market system, potentially causing damage to market participants, including investors.

Additionally, the requirement, under Rule 614(d)(9)(iv) and Rule 1004(c) of Regulation SCI, for a competing consolidator to conduct testing of its business continuity and disaster recovery plans with its designated participants and other industry SCI entities will help detect and improve the coordination of responses to system issues that could affect multiple market participants in the NMS stock market. This testing will help prevent these system disruptions from occurring and help reduce the severity of their effects, if they do occur.

Third, Rule 614(d)(9) and the addition of the definition of SCI competing consolidator to Regulation SCI will help ensure effective Commission oversight of new competing consolidators who are not currently SCI entities. Both Regulation SCI and Rule 614(d)(9)(iii)(C) will require a competing consolidator to notify the Commission[2331] and provide the Commission with updates if it experiences a systems disruption or systems intrusion that has more than a *de minimis* impact.[2332] Each quarter, an SCI competing consolidator will have to inform the Commission of

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[2331] Regulation SCI requires SCI entities to notify the Commission immediately upon any responsible SCI personnel having a reasonable basis to conclude that an SCI event has occurred. Similarly, Rule 614(d)(9)(iii)(C) requires competing consolidators to promptly notify the Commission upon responsible personnel having a reasonable basis to conclude that a system disruption or systems intrusion has occurred. The requirement for immediate or prompt notification, as applicable, does not apply to such events that a competing consolidator reasonably estimates would have no, or a *de minimis*, impact on the competing consolidator's operations or on market participants. See e.g., Rule 614(d)(9)(iii)(B)-(C); Regulation SCI Rule 1002(b)(5).

[2332] An SCI competing consolidator will be required to notify the Commission on Form SCI, and a competing consolidator that is not an SCI competing consolidator will be required to notify the Commission on Form CC. Additionally, each quarter SCI competing consolidators will be required to submit a report to the Commission on Form SCI of systems disruptions or systems intrusions that had no or a *de minimis* impact.

any planned material changes to its SCI systems and the security of indirect SCI systems. Each year an SCI competing consolidator also will have to provide the Commission with an SCI review of their compliance with Regulation SCI. This information will help ensure effective Commission oversight by enhancing the Commission's review of these competing consolidators and helping make the Commission aware of potential areas of weakness in the competing consolidator's systems that may pose risk to the entity or the market as a whole.

Additionally, the Commission believes that an exclusive SIP that becomes a competing consolidator may realize an incremental benefit relative to the baseline from lower SCI-related costs.[2333] Because the Commission assumes that enough competing consolidators will enter the market to provide for multiple viable sources of consolidated market data products,[2334] the Commission believes that an exclusive SIP will not need to incur the additional costs associated with being subject to the heightened requirements applicable to "critical SCI systems" if it chooses to operate a competing consolidator after the initial transition period.

(ii) Costs of Expanding Regulation SCI to Include Competing Consolidators

Competing consolidators will incur both direct and indirect compliance costs related to Rule 614(d)(9) and the addition of the definition of SCI competing consolidator to Regulation

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2333 The systems of the exclusive SIPs are "critical SCI systems" and are subject to heightened requirements. For example a critical SCI system needs to maintain backup systems that are designed to allow them to resume operations within two hours of a system outage (SCI entities only have the requirement to resume operations the day following a system outage). See Proposing Release, 85 FR at Section IV.B.2(f).

2334 See supra Section V.C.2(a) for a discussion of this assumption.

SCI.[2335] Although all competing consolidators will initially be subject to Rule 614(d)(9) during the initial transition period, the Commission believes that, after the transition period, many competing consolidators will be above the SCI competing consolidator threshold and eventually need to bear the higher costs Regulation SCI.[2336] Because Regulation SCI imposes some indirect requirements on other market participants interacting with SCI entities (e.g., vendors providing SCI systems to SCI entities), those market participants will also incur indirect costs from SCI competing consolidators.

Competing consolidators will incur initial and ongoing direct PRA and non-PRA compliance costs related to Rule 614(d)(9) and Regulation SCI. These costs will vary based on whether the competing consolidator is an SCI competing consolidator or whether it is subject to the provisions of Rule 614(d)(9).[2337]

The Commission believes that the 2018 estimates of initial PRA costs for new SCI entities and ongoing PRA costs for all SCI entities under Regulation SCI are largely applicable to SCI competing consolidators because the requirements are the same for all SCI entities and because the 2018 burden estimates were based on the Commission's experience over three years subsequent to Regulation SCI's adoption in 2014 including, for example, Commission staff's experience in conducting examinations of SCI entities and receiving and reviewing notifications

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2335 Direct compliance costs will include both costs that included in the PRA burden estimates as well as compliance costs that are not reflected in the PRA ("non-PRA"). See supra Section IV.D (for a discussion of the PRA burden estimates).

2336 See supra Section IV.G.3.

2337 See id.

and reports required by Regulation SCI.[2338] The 2018 SCI PRA Extension includes estimates distinguishing between new versus existing SCI entities. The Commission believes that, using the same new versus existing SCI entity framework, entrants that could become SCI competing consolidators can be divided into three groups: the existing exclusive SIPs; entrants that are existing SCI entities but with no direct experience operating in the consolidated market data business and needing to perform a new function with new SCI systems (e.g., a national securities association or national securities exchanges that do not currently operate an exclusive SIP); and entrants that are not currently subject to Regulation SCI (e.g., third-party aggregators that are not currently subject to Regulation SCI). The Commission estimates that the exclusive SIPs will not have any initial PRA costs related to Regulation SCI from becoming a competing consolidator because they are already SCI entities and would be operating a substantially similar business and performing a similar function in their role as competing consolidators.[2339] Because they would be entering an entirely new business and performing a new function with new SCI systems, SCI entities without direct experience operating in the consolidated market data business will each incur an initial PRA cost of approximately $326,000, which is approximately 50% of the Commission's initial cost estimates for an entirely new SCI entity.[2340] SCI competing

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2338 See supra note 1567. Two commenters stated that the Commission underestimated the costs of Regulation SCI in the Proposing Release. See IDS Letter I at 13 and STANY Letter II at 6–7. As discussed in detail above, the Commission disagrees with these commenters and believes it did not underestimate the costs associated with Regulation SCI. See supra Section IV.G.2.

2339 See supra note 1572 and accompanying text.

2340 These cost estimates are based on the 2018 SCI PRA Extension. See 2018 SCI PRA Extension, supra note 1567. See also supra Section IV.G discussing PRA burden

consolidators that are not currently subject to regulation SCI will each incur an initial PRA cost of approximately $625,000, which is the same estimated initial paperwork cost as those estimated for new SCI entities.[2341] The Commission estimates that all SCI competing consolidators will each incur ongoing annual PRA costs of approximately $804,000, which is the same as the ongoing costs for existing SCI entities estimated in the 2018 SCI PRA Extension.[2342]

Although the requirements of Rule 614(d)(9) are similar to some of the key provisions of Regulation SCI, Rule 614(d)(9) does not contain all of the provisions of Regulation SCI and will have lower compliance costs than Regulation SCI.[2343] For example, Rule 614(d)(9) does not contain a provision similar to the requirement for geographically diverse backup and recovery capabilities that is contained in Rule 1001(a)(2) of Regulation SCI. Therefore, the Commission estimates that the requirements of Rule 614(d)(9) will have initial and ongoing PRA costs that are approximately 33% of the PRA costs for compliance with all of the provisions of Regulation SCI.[2344] The Commission estimates that competing consolidators that are below the SCI competing consolidator threshold will each incur initial PRA costs of approximately $217,000 and ongoing annual PRA costs of approximately $268,000.

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estimates related to compliance with Regulation SCI and supra note 1573 and accompanying text.

2341 See supra note 1575 and accompanying text.

2342 See supra Section IV.G.

2343 See supra Section III.F.

2344 See supra note 1589 and accompanying text (discussing the PRA burden estimates for Rule 614(d)(9)).

As SCI entities, SCI competing consolidators will also incur direct non-PRA related compliance costs. In 2014, the Regulation SCI adopting release estimated that an SCI entity will incur an initial non-PRA cost of between approximately $320,000 and $2.4 million.[2345] Additionally, an SCI entity will incur an annual ongoing non-PRA cost of between approximately $213,600 and $1.6 million.[2346] The Commission believes that similar to the PRA cost estimates, these non-PRA related costs are also largely applicable to SCI competing consolidators. But the Commission is uncertain about the actual level of costs SCI competing consolidators will incur, because these costs could differ based on the type of potential entrant that becomes an SCI competing consolidator. The Commission believes that there are two reasons why SCI competing consolidators' non-PRA costs are likely to be on the lower end of these cost estimates.

First, these cost estimates include costs of having part of an SCI entity's system be a "critical SCI system," and therefore be subject to certain heightened resilience and information dissemination provisions of Regulation SCI. SCI competing consolidators' systems are not included within the scope of "critical SCI systems."[2347] The Commission believes that if SCI competing consolidators' systems are subject to the standard requirements of Regulation SCI, they will not have to incur compliance costs of the heightened requirements for "critical SCI systems." To the extent that the incremental costs of being subject to the heightened requirements for "critical SCI systems" versus the standard requirements for "SCI systems" is small, these cost savings will be low.

2345 SCI Adopting Release, supra note 1233, at nn.1943–44.

2346 Id. at nn.1945–46.

2347 See supra Section III.F.

Second, among all of the SCI entities, SCI competing consolidators have relatively simpler systems and fewer functions, and thus will have compliance costs closer to the lower end of the above non-PRA cost estimates. The above non-PRA cost estimates provide an average for all SCI entities, without distinguishing between different categories of SCI entities. However, the Regulation SCI adopting release explains that compliance costs will depend on the complexity of SCI entities' systems and they would be higher for SCI entities with more complex systems.[2348] SCI competing consolidators will likely have simpler systems and fewer functions relative to some of the other SCI entities, such as exchanges. As a result, the Commission believes that SCI competing consolidators' compliance costs are likely to be on the lower end of the average non-PRA cost estimates for all SCI entities.

Because Rule 614(d)(9) does not contain all of the provisions of Regulation SCI, the Commission believes that Rule 614(d)(9) will have lower initial and ongoing non-PRA compliance costs than Regulation SCI.[2349] Similar to the PRA cost estimates, the Commission estimates that the requirements of Rule 619(d)(9) will have initial and ongoing non-PRA costs that are approximately 33% of the non-PRA costs for compliance with all of the provisions of Regulation SCI. The Commission estimates that competing consolidators below the SCI competing consolidator threshold will each incur an initial non-PRA cost of between approximately $107,000 and $800,000.[2350] Additionally, competing consolidators below the SCI competing consolidator threshold will also each incur an annual ongoing non-PRA cost of between approximately $71,000 and $533,000. The Commission is uncertain about the actual

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2348 SCI Adopting Release, supra note 1233, at 634.

2349 See supra Section III.F.

2350 SCI Adopting Release, supra note 1233, at nn.1943–44.

level of costs competing consolidators below the SCI competing consolidator threshold will incur, because these costs could differ based on the state of the systems of the entrant that becomes a competing consolidator. Should a competing consolidator meet the threshold to become an SCI entity after the initial transition period, there would be additional costs at that time in order to comply with Regulation SCI, which will vary depending on the type of competing consolidator.[2351]

Additionally, the Commission believes that some competing consolidators' subscribers associated with the testing of business continuity and disaster recovery plans will incur Regulation SCI-related connectivity costs. Rule 1004 of Regulation SCI sets forth the requirements for testing an SCI entity's business continuity and disaster recovery plans with its designated members or participants. Rule 614(d)(9)(iv) requires competing consolidators that are not affiliated with exchanges that do not meet the threshold requirements for being an SCI competing consolidator to participate in the testing outlined in Rule 1004 of Regulation SCI. Competing consolidators and their designated subscribers would be subject to these same costs.[2352] The Regulation SCI adopting release estimated connectivity costs as part of these business continuity and disaster recovery plans to be approximately $10,000 per SCI entity member or participant.[2353] The Commission believes that these connectivity cost estimates will also be applicable to competing consolidators' designated subscribers.

2351 The Commission believes that the initial implementation costs for these entities to comply with Regulation SCI will approximately be the difference between their initial PRA and non-PRA costs under Rule 614(d)(9)(iv) and the initial PRA and non-PRA burdens based on their entity type, as described above in this section.

2352 See supra Section III.F.

2353 See SCI Adopting Release at n.2065.

The Commission believes that competing consolidators and various other market participants will incur certain indirect costs related to compliance requirements for SCI competing consolidators. The Commission believes that the costs to comply with Regulation SCI discussed above will also fall on third-party vendors employed by SCI competing consolidators to provide services used in their SCI systems. Regulation SCI requires that any system provided by a vendor to an SCI entity and used by that entity in its SCI system must also comply with Regulation SCI requirements. The Commission believes that all costs discussed above for competing consolidators to comply with Regulation SCI will also fall on third-party vendors employed by competing consolidators in the course of providing consolidated market data. Examples of such vendors may include communications firms employed by SCI competing consolidators to transport data from exchanges to the SCI competing consolidator's aggregation servers at various data centers. If many third-party vendors are employed by SCI competing consolidators in their consolidated market data business, the size of this cost may be significant.

Additionally, the Commission believes there is the potential for these costs to cause vendors to end certain existing business relationships with market participants who become SCI competing consolidators. It is possible that third-party vendors will not want to incur the costs that SCI competing consolidators may impose to assure that the SCI competing consolidator can comply with Regulation SCI requirements, and as a result be unwilling to provide services to the SCI competing consolidator's consolidated market data business. To the extent that this occurs, SCI competing consolidators will incur costs from having to find new vendors, form a new business relationship, and adapt their systems to the infrastructure of the new vendor. SCI competing consolidators may also elect to perform the relevant functions internally. To the

extent that SCI competing consolidators either find new vendors or perform the functions internally, it will represent an increased inefficiency in the market, since presumably the current market data vendors are the most efficient means of performing these functions.

The Commission believes that the technology supporting some of the services provided by vendors to current data aggregators (notably communications, such as microwave transmission) require significant expertise in order to be competitive and are difficult to replicate. To the extent this is the case, and to the extent that Regulation SCI requirements prevent SCI competing consolidators from using these vendors, the ability of SCI competing consolidators to provide consolidated market data in a manner that rivals current third-party aggregation practices may be significantly reduced.

(f) Economic Effects of the Decentralized Consolidation Model Pertaining to Self-Aggregators

As discussed above a number of market participants currently purchase proprietary data products from the exchanges and consolidate this data for their internal use or regulatory compliance.[2354] To permit self-aggregation under the decentralized consolidation model, the Commission defines a new type of market data user, self-aggregators.[2355]

Market participants that currently effectively self-aggregate and that decide to become self-aggregators under the decentralized consolidation model will have two choices. First, they may decide to limit the use of exchange data to the creation of consolidated market data, in

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2354 Some commenters agreed. See, e.g., MEMX Letter at 7; NYSE Letter II at 18. See also supra Section III.D.

2355 See supra Section III.D.2 for a definition of a self-aggregator.